

8.333 Recitation 6

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Convergence of Random Variables

What does it mean for one random variable to converge to another?

$$" \lim_{n \rightarrow \infty} X_n = Y "$$

1) **Pointwise convergence:** X_n, Y are functions $Y: \Omega \rightarrow \mathbb{R}$.

We can define pointwise convergence in the usual sense for functions:

$$\left| \text{Def } X_n \text{ converges pointwise to } Y \text{ if} \right. \\ \left. \lim_{n \rightarrow \infty} X_n(\omega) = Y(\omega) \quad \forall \omega \in \Omega \right.$$

This is the strongest kind of convergence - it is not useful, because it requires convergence even on measure zero sets.

2) **Almost sure convergence:** like pointwise, but it allows $\lim_{n \rightarrow \infty} X_n$ and Y to disagree on a set of measure zero.

$$\left| \text{Def } X_n \text{ converges almost surely to } Y \text{ if} \right. \\ \left. \mathbb{P} \left[\lim_{n \rightarrow \infty} X_n = Y \right] = 1. \right.$$

Equivalently:

$$\mathbb{P} \left[\left\{ \omega \in \Omega : \lim_{n \rightarrow \infty} X_n(\omega) \neq Y(\omega) \right\} \right] = 0.$$

Clearly (1) implies (2).

We typically write $X_n \xrightarrow{\text{a.s.}} Y$

This definition is much more useful but still too strong for some purposes (e.g. the CLT).

3) **Convergence in probability:** This says the probability of an unusual outcome falls as $n \rightarrow \infty$:

$$\left| \text{Def } X_n \text{ converges in probability to } Y \text{ if} \right. \\ \left. \lim_{n \rightarrow \infty} \mathbb{P}(|X_n - Y| > \varepsilon) = 0 \quad \forall \varepsilon > 0 \right.$$

4) Convergence in distribution (convergence in law, weak convergence):

Def X_n converges in distribution to Y if

$$\lim_{n \rightarrow \infty} P(X_n \in A) = P(Y \in A)$$

In ID: $\lim_{n \rightarrow \infty} F_{X_n}(x) = F_Y(x)$ (CDFs converge pointwise).

This is the most useful notion of convergence for us. The CLT is a statement about this.

How to prove convergence in distribution? The following imply convergence in dist:

- a) Pointwise convergence of CDF, or
- b) Pointwise convergence of probability density, or
- c) Pointwise convergence of characteristic functions, or
- d) Convergence of moments or cumulants.

It can be shown that $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4)$. The converse is not true in general, but there is a special case where $(4) \Rightarrow (3)$:

When Y is a constant y , convergence in distribution implies convergence in probability. We won't prove this, but it is plausible — dist approaches a delta fn.

With these definitions in mind, let us restate the CLT and LLN:

Let $\{X_i\}$ be a sequence of i.i.d RVs with finite moments.

Theorem: Central limit theorem:

The scaled sum $Y_N = \frac{1}{\sqrt{N}} \sum_{i=1}^N (X_i - \langle X \rangle)$ converges in distribution to a $\mathcal{N}(0, \langle X^2 \rangle_c)$ distribution.

$$Y_N \xrightarrow{\mathcal{D}} \mathcal{N}(0, \langle X^2 \rangle_c)$$

Proof: In lecture we showed that the cumulants of Y_N satisfy:

$$\langle Y_N \rangle = 0, \quad \langle Y_N^2 \rangle_c = \langle X^2 \rangle_c, \quad \lim_{N \rightarrow \infty} \langle Y_N^{k \geq 2} \rangle = 0.$$



Theorem: (Weak) Law of Large Numbers

The sample mean $\bar{X}_N \equiv \frac{1}{N} \sum_{i=1}^N X_i$ converges in probability to the true mean:

$$\bar{X}_N \xrightarrow{\mathcal{P}} \langle X \rangle$$

Proof Weak LLN follows from CLT:

$$\text{We know } \sqrt{N} (\bar{X}_N - \langle X \rangle) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \langle X^2 \rangle_c)$$

Using CLT is overkill
 $\rightarrow$ use Chebyshev's inequality

So

$$\begin{aligned} \mathbb{P}(|\bar{X}_N - \langle X \rangle| > \varepsilon) &= \mathbb{P}(\sqrt{N} |\bar{X}_N - \langle X \rangle| > \sqrt{N} \varepsilon) \\ &\propto \int_{\sqrt{N} \varepsilon}^{\infty} e^{-\frac{x^2}{2 \langle X^2 \rangle_c}} dx \xrightarrow{N \rightarrow \infty} 0. \quad \square \end{aligned}$$

Theorem: Strong Law of Large Numbers

$\bar{X}_N$ converges almost surely to $\langle X \rangle$.

We will not prove this here.

Weak law $\neq$ strong law: example, $X_n = \text{Bern}(1/n)$

Heads and tails: CLT and LDF regimes

Consider a sequence of independent $\text{Bern}(1/2)$ random variables X_n :

$$\mathbb{P}(X_n = x) = \begin{cases} 1/2 & \text{if } x = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

Let us define a new random variable, the sample mean:

$$\bar{X}_N \equiv \frac{1}{N} \sum_{k=1}^N X_k$$

$$\text{Mean: } \langle \bar{X} \rangle = \frac{1}{N} \sum_{k=1}^N \langle X_k \rangle = 1/2$$

We can write the distribution of $\bar{X}$ explicitly.

$$\begin{aligned} \mathbb{P}(N \bar{X}_N = k) &= \binom{N}{k} \left(\frac{1}{2}\right)^k \left(1 - \frac{1}{2}\right)^{N-k} = \binom{N}{k} 2^{-N}, \\ &= \mathbb{P}(\bar{X}_N = k/N) \quad k = 0, 1, 2, \dots \end{aligned}$$

$$\therefore \mathbb{P}(\bar{X}_N = r) = \binom{N}{Nr} 2^{-N} = \frac{N! 2^{-N}}{(Nr)! (N-Nr)!}, \quad r = 0, 1/N, 2/N, \dots$$

We can verify the two theorems:

1) Law of large numbers $\lim_{N \rightarrow \infty} \bar{X} \stackrel{P}{=} 1/2$.

2) CLT: $\sqrt{N} (\bar{X}_N - 1/2) \xrightarrow{D} \mathcal{N}(0, 1/4)$

1) LLN (roughly) $\text{Var}(\bar{X}_N) = \frac{1}{N^2} \sum_{k=1}^N \text{Var}(X_k) = \frac{1}{N} \frac{1}{4} \xrightarrow{N \rightarrow \infty} 0$

So $\bar{X}_N$ converges in distribution to a variable with variance zero.

A variable with variance zero is almost surely equal to its mean

$$\text{Var}(Y) = \mathbb{E}((Y - \langle Y \rangle)^2) = 0 \Rightarrow \mathbb{P}(Y = \langle Y \rangle) = 1$$

Can show more clearly using Chebyshev inequality.

2) CLT We will use cumulant generating functions $\Phi_X(k) = \log \hat{P}_X(k)$

Def $Y_N = \sqrt{N} (\bar{X}_N - 1/2)$ $= \log \langle e^{ikX} \rangle$

Recall cum gen fn of $\mathcal{N}(\mu, \sigma^2)$ is $i\mu t - \sigma^2 t^2/2$

So we want to show

$$\Phi_{Y_N}(k) \rightarrow -k^2/8$$

Indeed,

$$\begin{aligned} \langle e^{ikY_N} \rangle &= \langle e^{ik\sqrt{N}(\bar{X}_N - 1/2)} \rangle = e^{-ik\sqrt{N}/2} \langle e^{ik/\sqrt{N} \sum X_i} \rangle \\ &= e^{-ik\sqrt{N}/2} \langle e^{ik/\sqrt{N} X_1} \rangle^N \\ &= e^{-ik\sqrt{N}/2} \left(\frac{1}{2} e^{ik/\sqrt{N}} + \frac{1}{2} \right)^N \end{aligned}$$

$$\Phi_{Y_N}(k) = \log \langle e^{ikY_N} \rangle = -\frac{ik\sqrt{N}}{2} + N \log \left(e^{ik/\sqrt{N}} + 1 \right) - N \log 2$$

use $\log(1+z) = z - \frac{z^2}{2} + \mathcal{O}(z^3)$

Can show

$$\log \left(e^{ik/\sqrt{N}} + 1 \right) = \log 2 + \frac{ik}{2\sqrt{N}} - \frac{k^2}{8N} + \mathcal{O}\left(\left(\frac{k}{\sqrt{N}}\right)^3\right)$$

$$\therefore \Phi_{Y_N}(k) = -\frac{ik\sqrt{N}}{2} + \frac{ik\sqrt{N}}{2} - \frac{k^2}{8} + \mathcal{O}\left(\frac{k^3}{N^{1/2}}\right) \xrightarrow{N \rightarrow \infty} -k^2/8$$

The CLT gives good answers in the vicinity of the mean (distances of order $1/\sqrt{N}$ from the mean). It fails miserably far from the mean - for example, it would predict a +ve probability for -ve values of $\bar{X}$. Let's write the physicist's CLT for $\bar{X}$:

We know

$$\sqrt{N} (\bar{X}_N - 1/2) \xrightarrow{D} \mathcal{N}(0, 1/4)$$

This suggests

$$\sum_i X_i = N \bar{X}_N \stackrel{\text{approx, } N \gg 1}{\sim} \mathcal{N}\left(\frac{N}{2}, \frac{N}{4}\right)$$

What is $P(\bar{X}_N = 0)$?

$$\text{Exact: } P(\bar{X}_N = 0) = 2^{-N}$$

$$\text{CLT Approx: } P(\bar{X}_N \in (-0.5, 0.5)) = \sqrt{\frac{2}{N\pi}} \int_{-0.5}^{0.5} dx e^{-\frac{2(x-N/2)^2}{N}}$$

$$\text{For } N = 60, \quad \text{exact} \sim 10^{-19}$$

$$\text{approx} \sim 10^{-14}$$

Relative error
Gets worse for larger N .
Huge overestimate

Insurance companies are interested in probabilities of rare events!

How do we deal with rare events? *Large deviation theory.*

Let's apply Stirling's approx to exact distribution

$$p(k) = \frac{N! 2^{-N}}{(Nr)! (N-Nr)!}$$

$$\log p(k) = -N \log 2 + \log N! - \log(Nr!) - \log(N-Nr)!$$

$$\approx -N(\log 2 + r \log r + (1-r) \log(1-r)) \equiv -N I(r)$$

We write

$$p(k) \sim e^{-N I(r)}$$

" $\sim$ " means log equivalence:

$$A_N \sim B_N \text{ if } \lim_{N \rightarrow \infty} \frac{\log A_N}{\log B_N} = 1.$$

This is called a *large deviation form*

The function $I(r)$ is called a *rate function*. It is +ve and convex, with a unique zero at the mean $r=1/2$. Entropies are rate functions!

Lets use this now: $p(0) \sim e^{-N I(0)} \sim 10^{-19}$ much better!

Only very rarely can you compute rate function directly like this. LDF gives us ways to compute and deal with $I(r)$. To give you a flavor, I will state Cramer's theorem:

Let $x_1, x_2, \dots$ be a sequence of i.i.d with finite cumulant generating function $\phi(t) = \log \langle e^{tx_1} \rangle$ (here we use the real def, not complex).

Consider the sample mean

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N x_i$$

The $\bar{X}$ satisfies a large deviation principle

$$p(x) \sim e^{-N I(x)}$$

with $I(x) = \sup_{t \in \mathbb{R}} (tx - \phi(t)) = \phi^*(t)$

ϕ^* is the Legendre transform of ϕ .

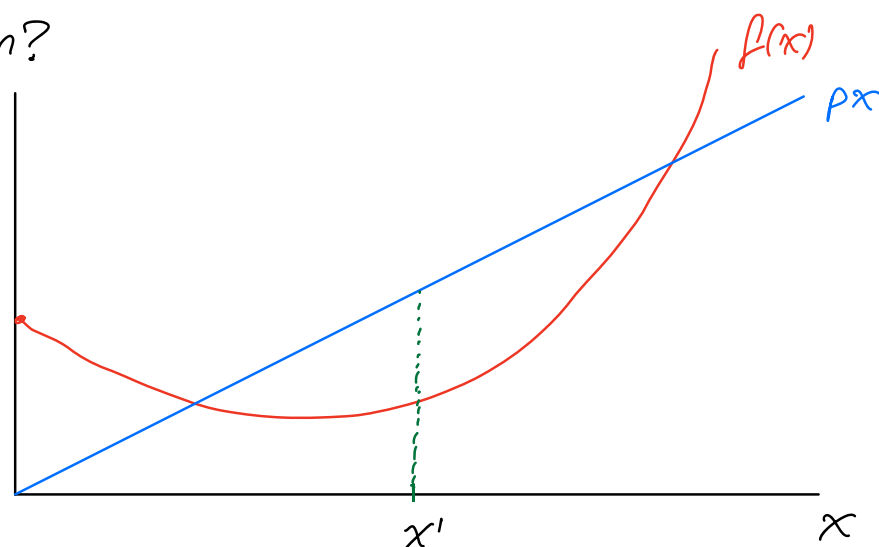
For more Large Deviation Theory in the context of Stat Mech, see review by Touchette.

Legendre transformation

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ a convex function. The Legendre transform $f^*: \mathbb{R} \rightarrow \mathbb{R}$ is the function defined by

$$f^*(p) = \sup_x (px - f(x))$$

What does this mean?



$$f^*(p) = px' - f(x')$$

p : conjugate variable

f^* : convex conjugate.

Notice

$$f^*(p) = \sup_x (px - f(x)) = (px - f(x)) \Big|_{x=(f')^{-1}(p)}$$

$$\text{because } \frac{d}{dx} (px - f(x)) = 0 \Rightarrow p - f'(x) = 0$$

Can check that this has the properties

$$1) f'(f^{*-1}(p)) = p$$

$$2) f^{**}(x) = f(x) \quad (\text{involutivity}).$$

Example: Classical Mechanics: $L(x, \dot{x}) = \frac{1}{2} m \dot{x}^2 - U(x)$

$$L^*(x, p) = \sup_{\dot{x}} (p \dot{x} - \frac{1}{2} m \dot{x}^2 + U(x))$$

$$= (p \dot{x} - \frac{1}{2} m \dot{x}^2 + U(x)) \Big|_{\dot{x} = (\partial_{\dot{x}} L)^{-1}(p)}$$

$$\partial_{\dot{x}} L(\dot{x}) = m \dot{x} \Rightarrow (\partial_{\dot{x}} L)^{-1}(p) = p/m$$

$$\therefore L^*(x, p) = \left(p \cdot \frac{p}{m} - \frac{1}{2} \frac{p^2}{m} + U(x) \right) = \frac{p^2}{2m} + U(x) = H(x)$$

Example: Thermodynamics: Consider the total energy $U(S, V)$.

Let us make a Legendre transform in S , with variable T :

$$U_T^*(T, V) = \sup_S (TS - U(S, V))$$

Notice that $U(S, V)$ is linear in S (extensivity), and by def $\frac{\partial U}{\partial S} = T$.

So $TS - U(S, V)$ is independent of S . Thus

$$U_T^*(T, V) = TS - U$$

The Helmholtz free energy is the negative of U_T^* :

$$F \equiv -U_T^* = TS - U.$$

If we do the same thing for Volume instead of entropy, we get
enthalpy $H = U + PV$, where $P = -\partial U / \partial V$.

If we Legendre transform w.r.t both V and S , get
Gibbs free energy: $G = U + PV - TS$.

The Isobaric Ensemble

In the derivation of the canonical ensemble, we considered a system and reservoir that can exchange energy and treated the composite system microcanonically.

We obtained $P(\psi) \propto e^{-\beta E(\psi)}$.

If we let both energy and volume fluctuate, we obtain the isobaric ensemble, where the Reservoir maintains a pressure and temperature but T, V fluctuate.

This leads to

$$p(\psi) \propto e^{-\beta E(\psi) - \beta P V(\psi)}$$

where $\beta P = \partial S / \partial V$.

This defines a "Gibbs Partition Function":

$$Z_I(T, P, N) = \sum_{\psi} e^{-\beta E(\psi) - \beta P V(\psi)} \quad *$$

And the Gibbs free energy:

$$G(N, T, P) = -k_B T \log Z_I$$

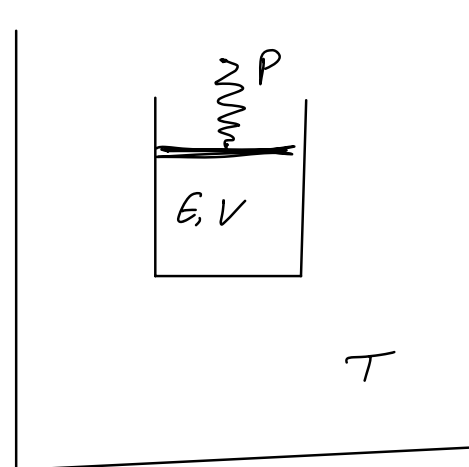
Using a saddle point on (*):

$$G = E - TS + PV.$$

The enthalpy is obtained as

$$H = \frac{-\log Z_I}{\partial \beta}$$

We have $\langle V \rangle = \partial G / \partial P$



$$\begin{aligned}
\mathbb{P}(|\bar{X}_n - 1/2| > \varepsilon) &= \mathbb{P}(|\sum_k (X_k - 1/2)| > N\varepsilon) \\
&\leq \mathbb{P}(\sum_k |X_k - 1/2| > N\varepsilon) \\
&= 1 - \mathbb{P}(\sum_k |X_k - 1/2| < N\varepsilon) \\
&\leq 1 - \mathbb{P}(|X_1 - 1/2| < N\varepsilon)^N
\end{aligned}$$

$$\mathbb{P}(\sum_k |X_k - 1/2| < N\varepsilon) \geq \mathbb{P}(|X_1 - 1/2| < N\varepsilon)^N$$